2.3 The Gelfand-Naimark-Segal Construction

The operational construction presented in Section **??** led to the conclusion that quantum observables can be embedded inside a unital non-commutative C^*-algebra. However, such an object carries no Hilbert space with it, so it is not obvious how this new formalism connects with the Dirac-Von Neumann axiomatization. The Gelfand-Naimark-Segal construction shows that every state on a C^*-algebra $\mathfrak{A}$ determines a Hilbert space $\mathcal{H}_\omega$ and a representation π_ω of $\mathfrak{A}$ as bounded operators $\mathcal{B}(\mathcal{H}_\omega)$, in which the state itself is reconstructed as the expectation value in a particular cyclic vector. This can be seen as the non-commutative counterpart of the Gelfand construction used in Chapter **??** for the classical case.

Before stating the definition of a representation, we isolate the class of maps between *-algebras that preserve all the algebraic relations.

Definition 2.3.1 ($*$-Homomorphism, $*$-isomorphism, $*$-automorphism)**.** *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two $*$-algebras with identity, as per Definition* **??**. *A $*$-homomorphism $\pi : \mathfrak{A} \to \mathfrak{B}$ is a map preserving all the algebraic relations of the $*$-algebra structure: It is linear, multiplicative and $*$-preserving, that is $\pi(A^*) = \pi(A)^*$ for every $A \in \mathfrak{A}$. A bijective $*$-homomorphism is called a $*$-isomorphism, and a $*$-isomorphism of $\mathfrak{A}$ onto itself is called a $*$-automorphism.*

A $*$-homomorphism gains importance once the target algebra $\mathfrak{B}$ is taken to be that of bounded operators on a Hilbert space.

Definition 2.3.2 (Representation, faithful and irreducible representation)**.** *Let $\mathfrak{A}$ be a C^*-algebra, as per Definition* **??***, and let $\mathcal{H}$ be a Hilbert space as in Definition* **??**. *A representation π of $\mathfrak{A}$ on $\mathcal{H}$ is a $*$-homomorphism, as per Definition 2.3.1, of $\mathfrak{A}$ into the C^*-algebra $\mathcal{B}(\mathcal{H})$ of bounded operators on $\mathcal{H}$. The representation π is called faithful when $\ker \pi = \{0\}$, and it is called irreducible when the only closed subspaces of $\mathcal{H}$ invariant under $\pi(\mathfrak{A})$ are $\{0\}$ and $\mathcal{H}$ itself.*

Having established representations, we return to the notion of cyclic vectors, introduced in Definition **??**. In fact, note that such vectors serves as a physical reference state from which the entire spectrum of accessible states is generated through the action of the observable algebra. Imposing cyclicity disconnects mathematical subspaces not physically accessible, as shown in the next remark.

Remark 2.3.3 (Cyclic vectors). Recolling Definition **??**, a vector $\Psi \in \mathcal{H}$ is called cyclic for a representation π when $\pi(\mathfrak{A})\Psi$ is dense in $\mathcal{H}$. In an irreducible representation, every non-zero vector is cyclic. In fact, the closure of $\pi(\mathfrak{A})\Psi$ is, by construction, a closed subspace invariant under $\pi(\mathfrak{A})$. Therefore, irreducibility forces it to coincide either with $\{0\}$ or with the whole of $\mathcal{H}$, but it cannot be $\{0\}$ because it contains at least $\Psi = \pi(\mathbb{1})\Psi$.

A representation π on a Hilbert space $\mathcal{H}$ together with a chosen cyclic vector Ψ is denoted by the triple $(\mathcal{H}, \pi, \Psi)$. Before presenting the key result of this section, we show that a $*$-homomorphism can only decrease norms, becoming norm preserving when it is injective.

Proposition 2.3.4. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two unital C^*-algebras as in Definition* **??** *and let $\pi : \mathfrak{A} \to \mathfrak{B}$ be a $*$-homomorphism as in Definition 2.3.1. Then*

$$\|\pi(A)\|_{\mathfrak{B}} \leq \|A\|_{\mathfrak{A}}, \qquad \forall A \in \mathfrak{A}. \tag{2.1}$$

If π is a $*$-isomorphism, equality holds.

The proof of this statement can be found in [1, Proposition 2.2.3]. The norm bound stated above indicates that any representation of a C^*-algebra must be realized by bounded operators. The theorem stated next, due to Gelfand and Naimark and later reformulated by Segal, shows that this quest always succeeds.

Theorem 2.1 (Gelfand-Naimark-Segal)**.** *Let $\mathfrak{A}$ be a unital C^*-algebra, as in Definition* **??***, and let ω be a state on $\mathfrak{A}$, as per Definition* **??**. *There exist a Hilbert space $\mathcal{H}_\omega$ and a representation $\pi_\omega : \mathfrak{A} \to \mathcal{B}(\mathcal{H}_\omega)$, as per Definition 2.3.2, such that*

i) $\mathcal{H}_\omega$ contains a cyclic vector Ψ_ω, as per Definition **??**, *ii)* $\omega(A) = \big(\Psi_\omega, \pi_\omega(A)\Psi_\omega\big)$ *for every* $A \in \mathfrak{A}$*,*

iii) every other representation $(\mathcal{H}, \pi, \Psi)$ *such that*

$$\omega(A) = \big(\Psi, \pi(A)\Psi\big), \qquad \forall A \in \mathfrak{A}, \tag{2.2}$$

is unitarily equivalent to π_ω*: There exists an isometry* $U : \mathcal{H}_\pi \to \mathcal{H}_\omega$ *such that*

$$U\pi(A)U^{-1} = \pi_\omega(A), \qquad U\Psi = \Psi_\omega. \tag{2.3}$$

Proof. Part i) The state ω turns the algebra $\mathfrak{A}$, regarded as a vector space, into a space with a semidefinite inner product, setting

$$(A, B) := \omega(A^*B), \qquad A, B \in \mathfrak{A}. \tag{2.4}$$

Positivity of ω yields $(A,A) = \omega(A^*A) \geq 0$ for every $A \in \mathfrak{A}$, while sesquilinearity of Equation (2.4) follows directly from antilinearity of ω and of the involution.
Consider the set

$$J := \{A \in \mathfrak{A} : \omega(B^*A) = 0,\ \forall B \in \mathfrak{A}\}. \tag{2.5}$$

By definition, $A \in J$ when A is orthogonal to the whole of $\mathfrak{A}$ with respect to $(\cdot,\cdot)$, and taking $B = A$ shows in particular that $(A,A) = \omega(A^*A) = 0$. That makes J the set of vanishing vectors of the semidefinite inner product as in Equation (2.4). We claim that J is a left ideal of $\mathfrak{A}$.

Recall that a left ideal of an algebra $\mathfrak{A}$ is a linear subspace $J \subseteq \mathfrak{A}$ that absorbs left multiplication by an arbitrary element of the algebra $\mathfrak{A}J \subseteq J$. That means that $CA \in J$ $\forall C \in \mathfrak{A}$ and $\forall A \in J$. To verify this property for the set in Equation (2.5), take $B, C \in \mathfrak{A}$ and $A \in J$. The Cauchy-Schwarz inequality associated with the semidefinite inner product from Equation (2.4) yields

$$|\omega(B^*(CA))| = |(B, CA)| = |(C^*B, A)| \leq (C^*B, C^*B)^{1/2}(A,A)^{1/2} = 0, \tag{2.6}$$

where the second equality regroups $\omega(B^*CA) = \omega\big((C^*B)^*A\big)$, while the right-hand side vanishes because $(A,A) = 0$, as $A \in J$. Since $B \in \mathfrak{A}$ is arbitrary, this shows $CA \in J$, which proves $\mathfrak{A}J \subseteq J$. Being a left ideal is the algebraic property that allows multiplication by a fixed element of $\mathfrak{A}$ to be defined consistently on equivalence classes modulo J.

We can therefore consider the quotient vector space $\mathfrak{A}/J$, whose elements are the equivalence classes $[A] = \{A + B : B \in J\}$, for $A \in \mathfrak{A}$. The inner product from Equation (2.4) descends to a well defined, strictly positive inner product on $\mathfrak{A}/J$. That is, because the value $\omega(A^*B)$ is unchanged when A or B is modified by an element of J, since J is the set of vanishing vectors identified above. It is strictly positive because a class $[A]$ with $([A],[A]) = 0$ has $A \in J$ by definition, that is, $[A] = [0]$. The completion of $\mathfrak{A}/J$ with respect to the norm induced by this inner product is a Hilbert space, denoted $\mathcal{H}_\omega$.

To each $A \in \mathfrak{A}$ we associate an operator denoted by $\pi_\omega(A)$ acting on $\mathcal{H}_\omega$, defined on the dense subspace $\mathfrak{A}/J$ by left multiplication,

$$\pi_\omega(A)[B] := [AB], \qquad [B] \in \mathfrak{A}/J. \tag{2.7}$$

If $[B] = [C]$, that is, $B - C \in J$, then $A(B - C) = AB - AC$ belongs to J as well, because J absorbs left multiplication by A. Hence $[AB] = [AC]$.

The operator $\pi_\omega(A)$ is bounded on $\mathfrak{A}/J$, with norm controlled by $\|A\|$. In fact, for every $B \in \mathfrak{A}$,

$$\|\pi_\omega(A)[B]\|^2 = \big([AB],[AB]\big) = \omega(B^*A^*AB) =: \omega_B(A^*A), \tag{2.8}$$

where $\omega_B(X) := \omega(B^*XB)$. The functional ω_B is itself positive and therefore continuous, as per [1, Prop. 1.6.3, Appendix C], and it satisfies $\omega_B(\mathbb{1}) = \omega(B^*B) = \|[B]\|^2$. Hence continuity of ω_B at the identity yields

$$\omega_B(A^*A) \leq \omega_B(\mathbb{1})\,\|A\|^2 = \|[B]\|^2\,\|A\|^2. \tag{2.9}$$

Therefore $\|\pi_\omega(A)[B]\| \leq \|A\|\,\|[B]\|$ for every $[B] \in \mathfrak{A}/J$. Hence $\pi_\omega(A)$ extends by continuity to a bounded operator on the whole of $\mathcal{H}_\omega$ by Theorem **??**, still denoted $\pi_\omega(A)$, with

$$\|\pi_\omega(A)\| \leq \|A\|. \tag{2.10}$$

Linearity of π_ω in A is immediate from Equation (2.7) and linearity of the product in $\mathfrak{A}$. Multiplicativity follows from associativity of the product, since

$$\pi_\omega(A)\pi_\omega(A')[B] = \pi_\omega(A)[A'B] = [AA'B] = \pi_\omega(AA')[B],$$

for all $A, A', B \in \mathfrak{A}$ The identity $\pi_\omega(A)^*[B] = \pi_\omega(A^*)[B]$ holds because $\big([AB],[C]\big) = \omega(B^*A^*C) = \big([B],[A^*C]\big)$ for all $B, C \in \mathfrak{A}$. Together with Equation (2.10), this shows that π_ω is a representation of $\mathfrak{A}$ on $\mathcal{H}_\omega$, as per Definition 2.3.2.

We now identify the vector $\Psi_\omega := [\mathbb{1}] \in \mathcal{H}_\omega$. By Equation (2.7), $\pi_\omega(A)\Psi_\omega = [A\mathbb{1}] = [A]$ for every $A \in \mathfrak{A}$, so the orbit $\pi_\omega(\mathfrak{A})\Psi_\omega$ coincides with the whole of $\mathfrak{A}/J$, which is dense in $\mathcal{H}_\omega$ by construction: Ψ_ω is a cyclic vector, proving part i).

Part ii) It follows from the same computation together with Equation (2.4) that

$$\big(\Psi_\omega, \pi_\omega(A)\Psi_\omega\big) = \big([\mathbb{1}],[A]\big) = \omega(\mathbb{1}^*A) = \omega(A). \tag{2.11}$$

Part iii) Let π be a representation on a Hilbert space $\mathcal{H}_\pi$ with a cyclic vector Ψ satisfying Equation (2.2). Define a map on the dense subspace $\pi_\omega(\mathfrak{A})\Psi_\omega = \mathfrak{A}/J$ of $\mathcal{H}_\omega$ by

$$U^{-1}\pi_\omega(A)\Psi_\omega := \pi(A)\Psi, \qquad A \in \mathfrak{A}. \tag{2.12}$$

This assignment is well defined on $\mathfrak{A}/J$. If $[A] = [A']$, that is, $A - A' \in J$, then in particular $\omega\big((A-A')^*(A-A')\big) = 0$. Taking $B = A–A'$ in Equation (2.5), and applying Equation (2.2) to the element $(A–A')^*(A–A') \in \mathfrak{A}$ yields

$$\big\|\pi(A)\Psi - \pi(A')\Psi\big\|^2 = \big(\Psi, \pi\big((A-A')^*(A-A')\big)\Psi\big) = \omega\big((A-A')^*(A-A')\big) = 0, \tag{2.13}$$

so $\pi(A)\Psi = \pi(A')\Psi$, as required. The same computation, applied to A^*A rather than to a difference, shows that U^{-1} is norm preserving on $\mathfrak{A}/J$,

$$\|\pi(A)\Psi\|^2 = \big(\Psi, \pi(A^*A)\Psi\big) = \omega(A^*A) = \big([A],[A]\big) = \|A\|^2. \tag{2.14}$$

Since Ψ is cyclic for π, the range of U^{-1} is dense in $\mathcal{H}_\pi$, and U^{-1} is an isometry defined on a dense subspace of $\mathcal{H}_\omega$. Therefore it extends uniquely to a unitary map $\mathcal{H}_\omega \to \mathcal{H}_\pi$, whose inverse is the required isometry $U : \mathcal{H}_\pi \to \mathcal{H}_\omega$. Equation (2.3) holds by construction on the dense subspace $\pi_\omega(\mathfrak{A})\Psi_\omega$, since

$$U^{-1}\pi_\omega(A')\pi_\omega(A)\Psi_\omega = U^{-1}\pi_\omega(A'A)\Psi_\omega = \pi(A'A)\Psi = \pi(A')\pi(A)\Psi = \pi(A')\,U^{-1}\pi_\omega(A)\Psi_\omega, \tag{2.15}$$

for all $A, A' \in \mathfrak{A}$. Furthermore it extends by continuity to the whole of $\mathcal{H}_\omega$. Taking $A = \mathbb{1}$ in the definition of U^{-1} yields $U^{-1}\Psi_\omega = \Psi$, that is, $U\Psi = \Psi_\omega$, which completes the proof. □

Remark 2.3.5 (Significance of the GNS construction). The GNS construction reduces the existence of Hilbert space representations of a C^*-algebra to that of states on it. That is guaranteed in general by [1, Prop. 1.6.4, Appendix C]. In addition, GNS Theorem 2.1 shows the dependence of two postulates of the Dirac-von Neumann axiomatization, namely that observables are represented by operators and that states are represented by vectors in a Hilbert space.

This theorem shows that cyclic representations are the basic building blocks of the representation theory of a C^*-algebra. In addition, the GNS construction of Theorem 2.1 is what produces it practically.

Remark 2.3.6 (Brief summary of the GNS construction). Given a state ω on a C^*-algebra $\mathfrak{A}$, as per Definition **??**, the proof of Theorem 2.1 produces the pair $(\mathcal{H}_\omega, \pi_\omega)$ through the following steps.

1. Define the sesquilinear form $(A, B) := \omega(A^*B)$ on $\mathfrak{A}$, positive semidefinite because ω is a state, as per Equation (2.4).

2. Its null space $J = \{A \in \mathfrak{A} : \omega(B^*A) = 0,\ \forall B \in \mathfrak{A}\}$, as per Equation (2.5), coincides with the set $\{A \in \mathfrak{A} : \omega(A^*A) = 0\}$ on account of the Cauchy-Schwarz inequality for $(\cdot,\cdot)$. Furthermore it is a left ideal of $\mathfrak{A}$.

3. The form $(\cdot,\cdot)$ descends to a inner product on the quotient $\mathfrak{A}/J$, whose completion is the Hilbert space $\mathcal{H}_\omega$.

4. The representation π_ω is defined on the dense subspace $\mathfrak{A}/J \subset \mathcal{H}_\omega$ by left multiplication, $\pi_\omega(A)[B] := [AB]$, as per Equation (2.7). It is extended to all of $\mathcal{H}_\omega$ by continuity.

5. The cyclic vector is $\Psi_\omega := [\mathbb{1}]$, and it reproduces the state, $\big(\Psi_\omega, \pi_\omega(A)\Psi_\omega\big) = \omega(A)$ for every $A \in \mathfrak{A}$.

Part iii) of Theorem 2.1 compares an arbitrary representation with the specific GNS representation of the state it reproduces. The same argument, applied twice, yields a direct comparison between any two cyclic representations that reproduce the same expectation values on their respective cyclic vectors.

Corollary 2.3.7. *Let π_1, π_2 be representations of a C^*-algebra $\mathfrak{A}$ on Hilbert spaces $\mathcal{H}_1, \mathcal{H}_2$, as per Definition 2.3.2, with cyclic vectors $\Psi_1 \in \mathcal{H}_1$, $\Psi_2 \in \mathcal{H}_2$ respectively, as in Definition* **??**. *If*

$$\big(\Psi_1, \pi_1(A)\Psi_1\big) = \big(\Psi_2, \pi_2(A)\Psi_2\big), \qquad \forall A \in \mathfrak{A}, \tag{2.16}$$

then π_1 and π_2 are equivalent, as per Definition **??**.

Proof. Denote by ω the common state $A \mapsto (\Psi_1, \pi_1(A)\Psi_1) = (\Psi_2, \pi_2(A)\Psi_2)$. By Theorem 2.1 iii), applied once to π_1 and once to π_2, both representations are equivalent to the GNS representation π_ω defined by ω. Composing the two unitaries obtained this way yields a unitary between $\mathcal{H}_1$ and $\mathcal{H}_2$ intertwining π_1 and π_2, so π_1 and π_2 are equivalent. □

Pure and Mixed States

The construction above produces many more states than the one it started from. In fact, every vector of $\mathcal{H}_\omega$, not only the distinguished cyclic vector Ψ_ω, determines a state on $\mathfrak{A}$ through the representation π_ω. This observation is specified by the following result.

Proposition 2.3.8. *Let ω be a state on a C^*-algebra $\mathfrak{A}$, as in Definition* **??**, *and let $(\mathcal{H}_\omega, \pi_\omega, \Psi_\omega)$ be its GNS representation, as per Theorem 2.1. For every $\Phi \in \mathcal{H}_\omega$, the functional $\varphi : \mathfrak{A} \to \mathbb{C}$ defined via*

$$\varphi(A) := \big(\Phi, \pi_\omega(A)\Phi\big), \tag{2.17}$$

where $A \in \mathfrak{A}$, is a positive linear functional on $\mathfrak{A}$.

Proof. The functional φ is positive because $\varphi(A^*A) = \|\pi_\omega(A)\Phi\|^2 \geq 0$ for every $A \in \mathfrak{A}$, and it is linear because π_ω is linear together with linearity of the inner product in its second argument. □

The next definition fixes the terminology used throughout the section to refer to such states.

Definition 2.3.9 (State vector)**.** *Let ω be a state on a C^*-algebra $\mathfrak{A}$, as in Definition* **??**, *and let $(\mathcal{H}_\omega, \pi_\omega, \Psi_\omega)$ be its GNS representation, as per Theorem 2.1. A vector $\Phi \in \mathcal{H}_\omega$ which defines a state as in Proposition 2.3.8 is called a state vector.*

Remark 2.3.10 (Physical Interpretation of State vectors)*.* Since, by the GNS construction, vectors of the form $\pi_\omega(A)\Psi_\omega$, for $A \in \mathfrak{A}$, are dense in $\mathcal{H}_\omega$, the state vectors obtained by acting with $\pi_\omega(\mathfrak{A})$ on Ψ_ω have the physical interpretation of states reached from ω through a physically realizable operation on the observables. If $\Phi \in \mathcal{H}_\omega$ is itself cyclic for π_ω, the representation $(\mathcal{H}_\omega, \pi_\omega, \Phi)$ is unitarily equivalent, as per Theorem 2.1 iii), to the GNS representation defined by φ.

In addition, distinct states on $\mathfrak{A}$ can be combined into new ones, simply by taking convex combinations of them. This operation identifies, among all states, those that describe a definite physical situation rather than a statistical combination of several ones.

Definition 2.3.11 (Mixture and pure state)**.** *Given two states ω_1, ω_2 on a C^*-algebra $\mathfrak{A}$, as in Definition* **??**, *the convex combination*

$$\omega := \lambda\omega_1 + (1-\lambda)\omega_2, \qquad 0 < \lambda < 1, \tag{2.18}$$

is again a state on $\mathfrak{A}$, called a mixture of ω_1 and ω_2, or a mixed state.

Alternatively, a state ω is called a pure state when it cannot be written as a convex combination Equation (2.18) of other states. These definitions correspond to those of the classical case expressed in Definition **??**. We can state an equivalent definition of pure states using the following operation.

Definition 2.3.12. *A state ω on a C^*-algebra $\mathfrak{A}$, as in Definition* **??**, *is said to majorize a positive linear functional ϕ on $\mathfrak{A}$ when $\omega - \phi$ is itself a positive linear functional.*

Using Definition 2.3.12, we can state equivalently that a state ω is called pure if the only positive linear functionals majorized by ω are of the form $\lambda\omega$, with $0 < \lambda < 1$.

The dichotomy between pure and mixed states has a counterpart at the level of GNS representations. In fact, purity of the state that generates the representation controls whether the representation itself is irreducible, as per Definition 2.3.2.

Proposition 2.3.13. *The GNS representation, as per Theorem 2.1, defined by a state ω on a C^*-algebra $\mathfrak{A}$ is irreducible if and only if ω is a pure state, as per Definition 2.3.11.*

For a complete proof see [1, Appendix D, Proposition 2.6.2]. A mixed state can therefore never be represented by a state vector, as per Definition 2.3.9, of an irreducible representation.

Mixed states admit a concrete description inside an irreducible representation. The next result specifies this construction. It shows that it always yields a positive linear functional on $\mathfrak{A}$.

Proposition 2.3.14. *Let π be an irreducible representation of a C^*-algebra $\mathfrak{A}$ on a Hilbert space $\mathcal{H}$ as in Definition 2.3.2. Given orthonormal vectors $\Psi_i \in \mathcal{H}$, $i = 1, \dots, N$, with $P_i \in \mathcal{B}(\mathcal{H})$ the corresponding one-dimensional orthogonal projections as in Definition* **??**, *let*

$$\rho := \sum_{i=1}^{N} \lambda_i P_i, \quad \textit{with} \quad \lambda_i \geq 0, \quad \sum_{i=1}^{N} \lambda_i = 1. \tag{2.19}$$

Then the functional

$$\omega_\rho(A) := \mathrm{Tr}\big(\rho\, \pi(A)\big), \qquad A \in \mathfrak{A}, \tag{2.20}$$

defines a state on $\mathfrak{A}$.

Proof. Complete the orthonormal set $\{\Psi_i\}_{i=1}^N$ to a complete orthonormal set $\{\Psi_n\}_{n=1}^d$ of $\mathcal{H}$, with $d \in \overline{\mathbb{R}}$ being the dimensions of $\mathcal{H}$. Expanding the trace in this basis, and using $\rho\Psi_n = \lambda_n \Psi_n$ for $n \leq N$ and $\rho\Psi_n = 0$ for $n > N$, we get

$$\omega_\rho(A) = \sum_{n=1}^{d} \big(\Psi_n, \rho\, \pi(A)\Psi_n\big) = \sum_{i=1}^{N} \lambda_i \big(\Psi_i, \pi(A)\Psi_i\big) = \sum_{i=1}^{N} \lambda_i\, \omega_i(A), \tag{2.21}$$

where $\omega_i(A) := (\Psi_i, \pi(A)\Psi_i)$ is, as per Definition 2.3.9, the state vector associated to Ψ_i. Being a convex combination of the states ω_i, the functional ω_ρ is itself a positive linear functional, and $\omega_\rho(\mathbb{1}) = \sum_i \lambda_i\, \omega_i(\mathbb{1}) = \sum_i \lambda_i = 1$: ω_ρ is a state on $\mathfrak{A}$. □

The operator ρ used in the previous construction is subject to the two conditions in Equation (2.19). These admit a characterization that does not refer to any particular choice of orthonormal vectors, stated in the next definition.

Definition 2.3.15 (Density matrix, folium of a representation). *An operator $\rho : \mathcal{H} \to \mathcal{H}$ on a Hilbert space $\mathcal{H}$, as in Equation* (2.19), *is called a density matrix when it is a positive trace class operator with* $\mathrm{Tr}\,\rho = 1$. *Given a representation π of a C^*-algebra $\mathfrak{A}$ on $\mathcal{H}$, the set of states of the form $\omega_\rho(A) = \mathrm{Tr}(\rho\, \pi(A))$, for ρ a density matrix on $\mathcal{H}$, is called the folium of the representation π.*

Remark 2.3.16. When ρ is a one-dimensional projection, the state ω_ρ of Proposition 2.3.14 coincides with the state vector, as per Definition 2.3.9, associated to the corresponding unit vector. Therefore, state vectors form a special case of the folium of Definition 2.3.15. When $N > 1$, ω_ρ is instead a mixed state, as per Definition 2.3.11, and generally every mixed state can be described by a density matrix.

The notion of faithfulness, already introduced at the level of representations in Definition 2.3.2, has a natural counterpart for states themselves. The two notions are directly related through the GNS construction.

Definition 2.3.17 (Faithful state). *A state ω on a C^*-algebra $\mathfrak{A}$ is called faithful when $\omega(A^*A) > 0$ for every non-zero $A \in \mathfrak{A}$.*

Remark 2.3.18. The GNS representation, as per Theorem 2.1, defined by a faithful state is itself faithful, as per Definition 2.3.2. Indeed, if $\pi_\omega(A) = 0$ then, in particular, $\pi_\omega(A)\Psi_\omega = [A] = [0]$, that is, A belongs to the ideal J of Equation (2.5), so $\omega(A^*A) = 0$, taking $B = A$ in Equation (2.5). Faithfulness of ω then forces $A = 0$, so $\ker \pi_\omega = \{0\}$.

Gelfand-Naimark Theorem for Observables

GNS representation, constructed in Theorem 2.1, can fail to be faithful. In fact, elements of $\mathfrak{A}$ that act differently as abstract algebra elements may be represented by the same, or even by the zero, operator. This would cause a failure of the separation property we presented in Definition **??**. Recovering a faithful realization of $\mathfrak{A}$ as an algebra of operators therefore requires working not with a single state but with a sufficiently large family of them at once. This is the content of the Gelfand-Naimark theorem, which shows that every C^*-algebra is isomorphic to an algebra of bounded operators on some Hilbert space.

Before proving that theorem, we need to state an equivalent definition of a separating state space. Recalling Definition **??**, a state set $\mathcal{S}$ is said to be *separating* when $\omega(A) = \omega(B)$ for all states $\omega \in \mathcal{S}$ implies $A = B$. By linearity and the fundamental properties of the states, this implies that $\omega(A^*A) = 0$ across all states if and only if $A = 0$. Hence, we can state an alternative operational definition more suitable for this framework.

Definition 2.3.19 (Separating property for a family of states). *Given a C^*-algebra $\mathfrak{A}$ as in Definition* **??**. *A family of states $\mathcal{F}$ is said to separate the algebra if, for every non-zero $A \in \mathfrak{A}$, there exists $\omega \in \mathcal{F}$ such that the represented operator $\pi_\omega(A)$ is not the zero operator.*

Using GNS Theorem 2.1, the operator $\pi_\omega(A)$ vanishes on $\mathcal{H}_\omega$ if and only if the element A lies in the Gelfand left ideal J defined in Equation (2.5), which is the condition $\omega(A^*A) = 0$. Consequently, the two definitions coincide. We are ready to present the non-commutative counterpart of Theorem **??**.

Theorem 2.2 (Gelfand-Naimark). *Every C^*-algebra $\mathfrak{A}$, as in Definition* **??***, is isomorphic to an algebra of bounded operators on a Hilbert space, as per Definition* **??**.

Proof. Using [1, Prop. 1.6.5, Appendix C] we can state the existence of a family $\mathcal{F}$ of separating states for every C^*-algebra. That is the state set described in Definition **??**. Fix one such family $\mathcal{F}$, and consider the direct sum of the corresponding GNS representations,

$$\mathcal{H} := \bigoplus_{\omega \in \mathcal{F}} \mathcal{H}_\omega, \qquad \pi := \bigoplus_{\omega \in \mathcal{F}} \pi_\omega. \tag{2.22}$$

Hence, $\mathcal{H}$ is the space of families $x = \{x_\omega\}_{\omega \in \mathcal{F}}$, with $x_\omega \in \mathcal{H}_\omega$ for every ω, such that $\sum_\omega \|x_\omega\|^2 < \infty$, equipped with the inner product $\big(\{x_\omega\}, \{y_\omega\}\big) := \sum_\omega (x_\omega, y_\omega)$, and π acts on such a family term by term, $[\pi(A)x]_\omega := \pi_\omega(A)x_\omega$.

The inner product on $\mathcal{H}$ is well defined and $\mathcal{H}$ is complete, by construction. In particular, convergence of $\sum_\omega (x_\omega, y_\omega)$ follows from the Cauchy-Schwarz inequality on each summand together with $\sum_\omega \|x_\omega\|^2 < \infty$ and $\sum_\omega \|y_\omega\|^2 < \infty$, and completeness of $\mathcal{H}$ is inherited coordinatewise from the completeness of each $\mathcal{H}_\omega$. The operator $\pi(A)$ is bounded on $\mathcal{H}$, since Proposition 2.3.4 yields $\|\pi_\omega(A)\| \le \|A\|$ for every $\omega \in \mathcal{F}$, that is

$$\|\pi(A)x\|^2 = \sum_{\omega \in \mathcal{F}} \|\pi_\omega(A)x_\omega\|^2 \le \|A\|^2 \sum_{\omega \in \mathcal{F}} \|x_\omega\|^2 = \|A\|^2 \|x\|^2. \tag{2.23}$$

Furthermore π inherits linearity, multiplicativity, and the $*$-preserving property from each π_ω, hence π is a representation of $\mathfrak{A}$ on $\mathcal{H}$, as per Definition 2.3.2.

Since $\mathcal{F}$ separates $\mathfrak{A}$, for every non-zero $A \in \mathfrak{A}$ there exists $\omega \in \mathcal{F}$ with $\pi_\omega(A) \neq 0$, hence $\pi(A) \neq 0$ and the representation π is faithful, $\ker \pi = \{0\}$, as per Definition 2.3.2, hence π is a $*$-isomorphism of $\mathfrak{A}$ onto the C^*-algebra $\pi(\mathfrak{A}) \subseteq \mathcal{B}(\mathcal{H})$. Applying Proposition 2.3.4 to the $*$-isomorphism π yields $\|\pi(A)\| = \|A\|$ for every $A \in \mathfrak{A}$. This implies that π is an isometric $*$-isomorphism of $\mathfrak{A}$ onto an algebra of bounded operators on the Hilbert space $\mathcal{H}$. $\square$

Remark 2.3.20 (Recovering the commutative case). When $\mathfrak{A}$ is commutative, Theorem 2.2 can be specialized to recover the Gelfand characterization of Chapter **??**. For a commutative C^*-algebra, irreducible representations coincide with the GNS representations defined by pure states, as per Proposition 2.3.13, and pure states on an commutative C^*-algebra are multiplicative, as per [1, Prop. 2.6.3]. Hence each such representation is one-dimensional, $\pi_\omega(A) = \omega(A)\mathbb{1}\ \forall A \in \mathfrak{A}$. Given a family $\mathcal{F}$ of pairwise inequivalent irreducible representations, it coincides with the Gelfand spectrum of $\mathfrak{A}$, as per Definition **??**, and the faithful representation

$$\pi(A) := \bigoplus_{\omega\in\mathcal{F}} \pi_\omega(A) \qquad \forall A \in \mathfrak{A}$$

is given by the collection $\{\omega(A) : \omega \in \mathcal{F}\}$ for every $A \in \mathfrak{A}$. That is because the correspondent continuous function of the observable A is given by $\omega(A)$, call it $\widetilde{A}$. The norm identity of Theorem 2.2 then reads $\|\widetilde{A}\|_\infty = \sup_\omega |\widetilde{A}(\omega)| = \sup_\omega |\omega(A)| = \|A\|$, and $\mathcal{F}$ is a compact Hausdorff space in the weak-$*$ topology, by the Alaoglu-Banach theorem, as per [2, Theorem IV.21], with each $\widetilde{A}$ continuous on it. This shows the structural difference between the commutative and the non-commutative case already announced in Section **??**.

2.4 The Weyl Algebra and its Representations

In the previous chapter, we characterized a commutative C^*-algebra as the correct algebraic model for a classical particle. The present section is devoted to find the algebra that embeds the canonical commutation relations between position and momentum in quantum mechanics. A naive answer, generating an algebra directly from the operators q and p, turns out to be incompatible with the very notion of a C^*-algebra. In fact, we show that no finite norm can be assigned to either operator once the commutation relations hold. Weyl's resolution replaces q and p with the bounded unitary exponentials that they formally generate. The resulting Weyl C^*-algebra becomes the correct object on which to invoke the noncommutative Gelfand-Naimark Theorem 2.2. In addition, von Neumann uniqueness theorem then shows that this representation, if it is regular, is unique up to unitary equivalence, so that the quantum particle possesses a single state space. The section closes by exhibiting the Schrödinger representation on $L^2(\mathbb{R})$. The following work in this section is based on [1, Chapter 3].

The Heisenberg relations and the boundedness obstruction

A classical particle in one space dimension is described by the algebra generated by the position q and the momentum p. The natural quantum analogue is obtained by requiring that the same two generators satisfy the following Heisenberg commutation relations.

$$[q, p] = i, \qquad [q, q] = 0, \qquad [p, p] = 0, \tag{2.24}$$

The value of $\hbar$ has been set to one by a suitable choice of units. Note that the discussion is restricted to a single space dimension for definiteness. The word *generated*, however, admits more than one interpretation, and this reading is specified the first task of the section.

Definition 2.4.1 (Heisenberg Lie algebra)**.** *The Heisenberg Lie algebra is the complex vector space spanned by q, p and the identity element* $\mathbb{1}$*, equipped with the Lie bracket determined by the commutation relations Equation* (2.24) *together with* $[q, \mathbb{1}] = [p, \mathbb{1}] = 0$.

The Lie bracket alone does not specify how q and p act as operators. The associative completion obtained in this way, rather than the Lie algebra of Definition 2.4.1 itself, is the structure usually meant with Heisenberg algebra. We need to examine the relations of this object to the operational framework of Section **??**.

Definition 2.4.2 (Heisenberg algebra)**.** *The Heisenberg algebra is the associative algebra generated, through sums and products, by q and p, with commutation relations fixed by Equation* (2.24).

The Heisenberg algebra of Definition 2.4.2 does not fit into the operational framework of Section **??**. The reason why is that q and p cannot be realized as bounded, or selfadjoint, elements of a C^*-algebra. The proposition below specifies this obstruction and quantifies how badly q and p fail to admit a finite C^*-norm.

Proposition 2.4.3. *Let q and p satisfy the Heisenberg commutation relations Equation* (2.24)*. If a C^*-norm could be assigned to q and p, then*

$$\|q\|\,\|p\| \geq \frac{n}{2}, \qquad \forall\, n \in \mathbb{N}. \tag{2.25}$$

Hence $\|q\|$ and $\|p\|$ cannot both finite.

Proof. The commutation relations Equation (2.24) entail, by induction on n,

$$[p, q^n] = -i\, n\, q^{n-1}. \tag{2.26}$$

If a C^*-norm existed on the algebra generated by q and p, submultiplicativity of the norm applied to Equation (2.26) would yield

$$n\, \|q^{n-1}\| \leq \|p\, q^n\| + \|q^n\, p\| \leq 2\, \|p\|\, \|q\|\, \|q^{n-1}\|. \tag{2.27}$$

Since $\|q^{n-1}\| = \|q\|^{n-1}$ by the C^*-identity and since it cannot vanish for any n. We are contradicting Equation (2.24). Dividing by $\|q^{n-1}\|$ therefore yields Equation (2.25) for every $n \in \mathbb{N}$, and letting n grow without bound shows that $\|q\|$ and $\|p\|$ cannot both be finite. □

Remark 2.4.4. The mathematical obstruction identified in Proposition 2.4.3 has a direct operational counterpart. Since any experimental apparatus is subject to scale bounds, what is actually measured is never q or p themselves rather bounded functions of them. Among this we include the position within the region accessible to the apparatus and the momentum within an interval set by the available energy. A formulation built on the Heisenberg algebra of Definition 2.4.2 therefore involves an extrapolation beyond what is operationally accessible in the sense of Section **??**. Such an extrapolation that carries no physical consequence yet forces a change of strategy at the algebraic level.

Proposition 2.4.3 is, in fact, an elementary instance of a considerably more general fact. This result makes no reference to the specific value of the commutator between q and p and holds for an arbitrary pair of bounded operators.

Theorem 2.3 (Wintner's Obstruction Theorem)**.** *Let $\mathcal{H}$ be a Hilbert space, as in Definition* **??***, and let $A, B \in \mathcal{B}(\mathcal{H})$. Then*

$$AB - BA \neq c\,\mathbb{1}, \qquad \forall\, c \in \mathbb{C} \setminus \{0\}. \tag{2.28}$$

The result was first proved by Wintner, with an independent proof given shortly afterwards by Wielandt. The proof is not reproduced here, and the reader is referred to [3, Proposition 11.32]. Taking $A = q$ and $B = p$ in Theorem 2.3 recovers the impossibility already established by direct computation in Proposition 2.4.3.

The Weyl Operators and the Weyl Algebra

Weyl's proposal replaces the problematic extrapolation with a family of operators that are bounded by construction, namely the formal exponentials of q and p. In addition, it is on these that a C^*-algebra can be built.

Remark 2.4.5. Note from from now on, $s \in \mathbb{N}$ will denote a finite number of dimensions.

Definition 2.4.6 (Weyl operators)**.** *Given q and p satisfying the Heisenberg commutation relations Equation* (2.24) *on $\mathbb{R}^s$, the Weyl operators are the bounded formal functions*

$$U(\alpha) = e^{i\alpha q}, \qquad V(\beta) = e^{i\beta p}, \qquad \alpha, \beta \in \mathbb{R}^s. \tag{2.29}$$

The algebraic properties of the Weyl operators can be read off from their formal relation to q and p, through an identity that will be used again in the proof of the von Neumann uniqueness theorem below.

Lemma 2.4.7 (Baker–Hausdorff formula)**.** *Let A, B be densely defined operators on a Hilbert space, as in Definition* **??***, with a common dense domain on which A, B, $A+B$ and their exponentials can be freely applied. Suppose that $C := [A, B]$ commutes with both A and B. Then*

$$e^A\, e^B = e^{A+B+[A,B]/2}. \tag{2.30}$$

A complete proof of this lemma can be found in [1, p.61]. Equation (2.30) applies to q and p as soon as the commutator between the two exponents is a scalar. That is the situation encompassed by the Heisenberg relations (2.24).

Proposition 2.4.8 (Weyl relations)**.** *The Weyl operators of Definition 2.4.6 satisfy*

$$U(\alpha)\,V(\beta) = V(\beta)\,U(\alpha)\,e^{-i\alpha\beta}, \tag{2.31}$$

$$U(\alpha)\,U(\alpha') = U(\alpha+\alpha'), \qquad V(\beta)\,V(\beta') = V(\beta+\beta'). \tag{2.32}$$

Proof. The element $[i\alpha q, i\beta p] = -i\alpha\beta$ is a scalar multiple of the identity by Equation (2.24), hence commutes with both $i\alpha q$ and $i\beta p$, and Lemma 2.4.7 applies to the pair $A = i\alpha q$, $B = i\beta p$. It yields

$$U(\alpha)\,V(\beta) = e^{i\alpha q + i\beta p - i\alpha\beta/2}. \tag{2.33}$$

Exchanging the roles of A and B yields, by the same argument,

$$V(\beta)\,U(\alpha) = e^{i\alpha q + i\beta p + i\alpha\beta/2} = U(\alpha)\,V(\beta)\,e^{i\alpha\beta}, \tag{2.34}$$

which is Equation (2.31). Equations (2.32) follow from Equation (2.24), since q commutes with itself and p with itself. Hence no correction term of the form $[A,B]/2$ appears in Lemma 2.4.7 when both operators belong to the same one-parameter family. □

The relations of Proposition 2.4.8, referred to as the Weyl form of the canonical commutation relations, no longer make reference to unbounded operators. It can therefore be taken as the defining relations of an abstract algebra.

Definition 2.4.9 (Weyl algebra)**.** *The Weyl algebra $\mathfrak{A}_W$ is the abstract algebra generated, through complex linear combinations and products, by abstract elements $U(\alpha)$, $V(\beta)$ with $\alpha, \beta \in \mathbb{R}$, subject to the relations Equation* (2.31) *and Equation* (2.32)*.*

Remark 2.4.10. A natural involution on $\mathfrak{A}_W$ is fixed by declaring $U(\alpha)^* = U(-\alpha)$ and $V(\beta)^* = V(-\beta)$, matching the selfadjointness that q and p would carry at the formal level. The Weyl relations then force $U(\alpha)$ and $V(\beta)$ to be unitary, and consequently every monomial in U and V has unit norm as well, since the Weyl relations reduce any such monomial to a single product $U(\alpha)V(\beta)$ up to a phase.

Assigning a norm to the remaining elements of $\mathfrak{A}_W$, so as to obtain a C^*-algebra, is a nontrivial step, shown in the following result.

Proposition 2.4.11. *There exists a unique C^*-norm on $\mathfrak{A}_W$, and the completion of $\mathfrak{A}_W$ with respect to it is a C^*-algebra.*

This proposition is not proved here, and we refer the reader to [4, Theorem 5.2.8]. With a unique C^*-norm available on $\mathfrak{A}_W$, its completion can now be regarded as a C^*-algebra. The next definition fixes the notation for this completed algebra, the object to which the noncommutative Gelfand-Naimark theorem will be applied below.

Definition 2.4.12 (Weyl C^*-algebra)**.** *The completion of $\mathfrak{A}_W$ with respect to the norm of Proposition 2.4.11 is a C^*-algebra, still denoted $\mathfrak{A}_W$, called the Weyl C^*-algebra.*

The quantum particle is defined as the quantum system characterized by the Weyl C^*-algebra $\mathfrak{A}_W$.

Remark 2.4.13. The states of the quantum particle are identified with the representations of $\mathfrak{A}_W$, as the states of the classical particle were identified in Section **??** with the representations of the corresponding commutative algebra. What Theorem 2.2 does not by itself provide is uniqueness, since a C^*-algebra can admit inequivalent representations, and it is this question that occupies the remainder of the section.

Before turning to uniqueness, it is worth recording the structure implicit in the Weyl relations, since it offers an equivalent description of representations of $\mathfrak{A}_W$.

Definition 2.4.14 (Heisenberg Lie group). *The Heisenberg Lie group is the Lie group whose elements are triples $(\alpha, \beta, \lambda) \in \mathbb{R}^s \times \mathbb{R}^s \times \mathbb{R}$, with group law*

$$(\alpha, \beta, \lambda)(\alpha', \beta', \lambda') = \Big(\alpha + \alpha',\ \beta + \beta',\ \lambda + \lambda' + \tfrac{1}{2}(\alpha'\beta - \alpha\beta')\Big). \tag{2.35}$$

Remark 2.4.15. The group law Equation (2.35) is obtained from the identification $(\alpha, \beta, \lambda) \longleftrightarrow \exp i(\alpha q + \beta p + \lambda \mathbb{1})$ together with the Weyl relations of Proposition 2.4.8. It exhibits the Heisenberg Lie group as the Lie group associated, through the exponential map, with the Heisenberg Lie algebra of Definition 2.4.1. Determining the representations of the Weyl C^*-algebra $\mathfrak{A}_W$ is therefore equivalent to determining the unitary representations of the Heisenberg Lie group.

Regularity and the von Neumann Uniqueness Theorem

Theorem 2.2 guarantees the existence at least one representation of $\mathfrak{A}_W$ on a Hilbert space. Yet it carries no informations about how many inequivalent ones exist. An infinite multiplicity of inequivalent representations would mean that the algebraic characterization of the quantum particle given in Definition 2.4.12 fails to determine its physics uniquely. That is this possibility that the regularity condition introduced below is planned to rule out.

Definition 2.4.16 (Regular representation). *A representation π of the Weyl algebra $\mathfrak{A}_W$, as in Definition 2.4.9, on a separable Hilbert space $\mathcal{H}$, as in Definition* **??**, *is regular if $\pi(U(\alpha))$ and $\pi(V(\beta))$ are strongly continuous one-parameter groups of unitary operators in α and β, respectively as per Definition* **??**.

Regularity is useful because strongly continuous one-parameter unitary groups admit self-adjoint generators, by the following theorem.

Theorem 2.4 (Stone's theorem). *Let $t \mapsto U(t)$ be a strongly continuous one-parameter group of unitary operators on a Hilbert space $\mathcal{H}$, as per Definition* **??**. *Then there exists a unique selfadjoint operator A on $\mathcal{H}$, generally unbounded and densely defined as per Definition* **??**, *such that*

$$U(t) = e^{itA}, \qquad \forall\, t \in \mathbb{R}. \tag{2.36}$$

Conversely, every self-adjoint operator A on $\mathcal{H}$ generates in this way a strongly continuous one-parameter unitary group.

The operator A is called the generator of U. The proof combines the spectral theorem for self-adjoint operators with the properties of strongly continuous unitary groups. The reader is referred to [3, Theorem 9.33] for a complete textbook treatment.

Remark 2.4.17. Although regularity is formally defined by requiring the strong continuity of the one-parameter unitary groups $\pi(U(\alpha))$ and $\pi(V(\beta))$, it constitutes a mild condition that is satisfied in non pathological physical models. For families of unitary operators, strong continuity is equivalent to the weak one. This is a consequence of the following identity.

$$||(U(t) - \mathbb{1})\Psi||^2 = 2||\Psi||^2 - 2\,\mathrm{Re}\,(\Psi, U(t)\Psi). \tag{2.37}$$

That demonstrates that the strong limit exists provided the real part of the transition amplitude $(\Psi, U(t)\Psi)$ approaches $||\Psi||^2$ as $t \to 0$. That is the weak continuity condition. Furthermore, on a separable Hilbert space, weak continuity is strictly equivalent to weak measurability of the scalar mappings $t \mapsto (\Phi, U(t)\Psi)$. Therefore, verifying regularity reduces to checking the Lebesgue measurability of these matrix elements $(\Psi, U(t)\Psi)$.

A key physical consequence of regularity concerns the unbounded observables. By Stone's theorem, strong continuity is the hypothesis for one-parameter unitary groups to admit unique self-adjoint generators on the Hilbert space $\mathcal{H}$. These generators, which are unbounded, correspond to the momentum and position operators, p and q, within the representation π. Regularity also shows that these operators share a common, dense, invariant domain. Therefore, this condition provides the framework necessary to reconstruct the foundational unbounded observables of the quantum particle.

Having established regularity, we can state the next theorem to classify representations.

Theorem 2.5 (von Neumann uniqueness theorem)**.** *All the regular irreducible representations of the Weyl C^*-algebra $\mathfrak{A}_W$ are unitarily equivalent.*

A complete proof of this statement can be found in [1, Theorem 3.2.2].

Remark 2.4.18 (The Fock state and unbounded generators). The constructive argument of the proof identifies a distinguished state on $\mathfrak{A}_W$, known as the *Fock state* ω_F. It is unambiguously defined by its expectation values on the generating elements of the algebra in the following way

$$\omega_F\left(e^{i\alpha\beta/2}\,U(\alpha)\,V(\beta)\right) = e^{-(|\alpha|^2+|\beta|^2)/4}. \tag{2.38}$$

The GNS construction associated with ω_F yields the *Fock representation*, and its corresponding cyclic vector Ψ_0 is called the *Fock vector*. By exploiting the regularity of the representation, differentiating Equation (2.38) at $\alpha = \beta = 0$ shows that Ψ_0 satisfies

$$(q + ip)\Psi_0 = 0.$$

This is an unbounded equation obtained from a statement that only ever involved bounded operators, providing an instance of the mechanism described earlier. The unbounded operators q and p act on Ψ_0 although they are not themselves elements of $\mathfrak{A}_W$. Their action is recovered by differentiating the bounded representation along the directions that regularity keeps under control.

The Schrödinger representation

Theorem 2.5 reduces the classification of regular irreducible representations to the exhibition of a single example. Historically, the standard choice is the following.

Definition 2.4.19 (Schrödinger representation)**.** *The Schrödinger representation π_S of the Weyl C^*-algebra $\mathfrak{A}_W$, as per Definition 2.4.9, acts on the Hilbert space $\mathcal{H} = L^2(\mathbb{R}^s, d^s x)$ by*

$$(U(\alpha)\psi)(x) = e^{i\alpha x}\,\psi(x), \qquad (V(\beta)\psi)(x) = \psi(x+\beta), \qquad \psi \in \mathcal{H}. \tag{2.39}$$

The definition would be of little use if it did not satisfy the two hypotheses of Theorem 2.5, regularity and irreducibility, and both properties are verified from Equation (2.39).

Proposition 2.4.20. *The Schrödinger representation of Definition 2.4.19 is a regular, irreducible representation of $\mathfrak{A}_W$.*

A full proof of this proposition can be found in [1, p.65].

Remark 2.4.21. The physical states of a quantum particle are, in this representation, the vectors of $L^2(\mathbb{R}^s)$, usually called Schrödinger wave functions. By Stone's Theorem 2.4, the strong continuity of $U(\alpha)$ and $V(\beta)$ established in Proposition 2.4.20 guarantees the existence, as strong limits on a dense domain, of the selfadjoint generators

$$(q\psi)(x) = x\,\psi(x), \qquad \psi \in D_q = \{\psi \in L^2(\mathbb{R}) : x\psi \in L^2(\mathbb{R})\}, \tag{2.40}$$

$$(p\psi)(x) = -i\,\frac{d\psi}{dx}(x), \qquad \psi \in D_p = \{\psi \in L^2(\mathbb{R}) : \psi' \in L^2(\mathbb{R})\}, \tag{2.41}$$

The derivative in Equation (2.41) is understood in the weak sense. The Schwartz space $\mathcal{S} := \mathcal{S}(\mathbb{R}^s)$ of smooth functions of fast decrease is a common dense domain of essential selfadjointness for q and p, following [2, Chapter VIII, Section 2]. The Heisenberg relations Equation (2.24) hold on $\mathcal{S}$.

Equations (2.40) and (2.41) are the concrete, unbounded realization of the operators that Proposition 2.4.3 excluded from the Weyl C^*-algebra $\mathfrak{A}_W$ from the outset. In fact, theorem 2.4 provides a reconstruction of q and p as the infinitesimal generators, as per Definition **??**, of the bounded one-parameter groups $\pi_S(U(\alpha))$ and $\pi_S(V(\beta))$.

2.5 The Dirac-von Neumann Axioms Revisited

The reconstruction carried out in Section **??** and Section 2.3 started from purely operational data, namely preparations and tests. Consequently, we show the embedding of Definition **??** into a unital C^*-algebra $\mathfrak{A}$. Historically, however, quantum mechanics reached this level of rigor along the opposite route. Heisenberg's matrix mechanics of 1925 was cast by Paul Dirac into a first systematic formulation set directly at the level of a Hilbert space. Von Neumann later refined it into the set of postulates that most textbooks still adopt today. Therefore, it is worth recording this historical formulation and matching each of its postulates against the results already established above. The comparison shows that every Dirac-von Neumann axiom finds a counterpart among the theorems proved earlier in the chapter. Following [1, Sect. 3.6], we state the axioms of the Hilbert space formulation of quantum mechanics.

Definition 2.5.1 (Dirac-von Neumann axioms)**.** *The Dirac-von Neumann formulation of quantum mechanics rests on the following postulates.*

I. States. *The states of a quantum system are described by density matrices $\rho = \sum_i \lambda_i P_i$, as in Definition 2.3.15, with $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$, and P_i one-dimensional projections, in a separable complex Hilbert space $\mathcal{H}$, as in Definition* **??**.

II. Observables. *The observables of a quantum system are described by the set of bounded self-adjoint operators in $\mathcal{H}$, as in Definition* **??**.

III. Expectations. *If a state ω is described by the vector $\Psi_\omega \in \mathcal{H}$, the expectation value of an observable A is the Hilbert space matrix element $\omega(A) = (\Psi_\omega, A\Psi_\omega)$, or more generally $\omega(A) = \mathrm{Tr}(\rho_\omega A)$ for a state described by the density matrix ρ_ω.*

IV. Canonical quantization. *For a system with finitely many degrees of freedom, the canonical coordinates and momenta satisfy $[q_i, q_j] = 0 = [p_i, p_j]$ and $[q_i, p_j] = i\hbar\delta_{ij}$, as in Equation* (2.24).

V. Schrödinger representation. *The commutation relations of Axiom IV are represented on $\mathcal{H} = L^2(\mathbb{R}^s, dx)$ by $(q_i\psi)(x) = x_i\psi(x)$ and $(p_j\psi)(x) = -i\hbar\,\partial\psi/\partial x_j(x)$.*

The five constructions that follow take up each axiom in the order in which Definition 2.5.1 lists them. We compare it against the material already developed in this chapter.

Recovering Axiom I. The state space S of Definition **??**, together with the identification of pure and mixed states of Definition 2.3.11, plays the role of Axiom I. A pure state is represented, through the GNS construction of Theorem 2.1, by a unit vector in the Hilbert space $\mathcal{H}_\omega$, since Proposition 2.3.13 identifies purity of ω with irreducibility of the associated representation π_ω. A mixed state is instead represented by a density matrix on $\mathcal{H}_\omega$ as per Definition 2.3.15, and Proposition 2.3.14 shows that every such density matrix defines a state. Separability of $\mathcal{H}_\omega$, required by Axiom I, is not automatic in this construction, and the next subsection discusses the circumstances under which it can fail. However, in most of physical systems, this hypothesis can be verified.

Recovering Axiom II The embedding of Definition **??** identifies the observable set $\mathcal{O}$ of Definition **??** with the selfadjoint sector $\mathfrak{A}_{sa}$ of the ambient C^*-algebra, matching Axiom II directly. Boundedness of the observables, absent from Dirac's original formulation according to [1, Sect. 3.6], is not an additional postulate in the present reconstruction. It descends from Definition **??** that assigns a finite norm to every observable. In fact, in the previous subsection, we showed that even unbounded operators like position and momentum can be represented as bounded operators using Weyl's algebra of Definition 2.4.9.
The passage from an abstract state on a C^*-algebra to a Hilbert space matrix element, postulated outright in Axiom III, is instead a theorem of the operational construction.

Corollary 2.5.2 (Expectation values from the GNS construction)**.** *Let ω be a state on a C^*-algebra $\mathfrak{A}$ as per Definition* **??***, and let $(\mathcal{H}_\omega, \pi_\omega, \Psi_\omega)$ be its GNS representation as per Theorem 2.1. For every $A \in \mathfrak{A}$,*

$$\omega(A) = (\Psi_\omega, \pi_\omega(A)\Psi_\omega). \tag{2.42}$$

More generally, for every density matrix ρ on $\mathcal{H}_\omega$ as per Definition 2.3.15, the state ω_ρ of Proposition 2.3.14 satisfies $\omega_\rho(A) = \mathrm{Tr}(\rho\, \pi_\omega(A))$.

Proof. Equation (2.42) is part ii) of Theorem 2.1. The general statement is Equation (2.20) of Proposition 2.3.14, applied to the representation π_ω. □

The last two axioms stand apart from the first three. The axioms I to III describe general mathematical structures common to every quantum system while IV and V refer to one specific realization. The remaining two remarks take up this beginning with the commutation relations imposed by Axiom IV.

Recovering Axiom IV. The commutation relations of Axiom IV are Equation (2.24) of Definition 2.4.1. They identify a single instance among the class of Definition **??**. That is the quantum particle whose algebra of observables is the Weyl C^*-algebra $\mathfrak{A}_W$ of Definition 2.4.12.

Recovering Axiom V. The Schrödinger representation of Axiom V coincides with Definition 2.4.19. Axiom V in Definition 2.5.1, however, postulates the usage of this representation, while, in the algebraic reconstruction, the von Neumann uniqueness Theorem 2.5, shows that it is, up to unitary equivalence, the only regular irreducible representation of $\mathfrak{A}_W$. Note that the restriction to a finite number s of dimensions, introduced ahead of Definition 2.4.6, is an hypothesis of the uniqueness theorem itself. In the next subsection, it is discussed what becomes of Axiom V once this hypothesis is dropped.

Limits of the GNS Construction and the Infinite-Dimensional Case

The construction in Section 2.5 isolated the hypothesis on which the whole uniqueness statement of Theorem 2.5 rests. This restriction to a finite number of degrees of freedom is the boundary between the general postulates I to III and the special postulates IV and V. Note that theories like quantum field theory, quantum statistical mechanics, and quantum many-body systems all require infinitely many degrees of freedom. In fact, the regime in which the Schrödinger representation of Definition 2.4.19 ceases to be available, see [1, Sect. 3.6].

Remark 2.5.3 (Inequivalent representations). The GNS construction of Theorem 2.1 remains valid without modification for the algebra of a field theory or of an infinite spin chain. In the infinite-dimensional analogue of the Weyl algebra of Definition 2.4.9, regular irreducible representations as per Definition 2.4.16 are no longer unitarily equivalent to one another. In particular, different equilibrium temperatures or different values of a mass parameter, correspond to physically distinct situations rather than to a single Hilbert space picture.

A second, more technical mismatch concerns the separability required by Axiom I itself. Remark 2.5 already presented this point. It is worth stating how the algebraic construction can fail to prove it.

Remark 2.5.4 (Separability). When $\mathfrak{A}$ is itself separable in the norm topology, the GNS space $\mathcal{H}_\omega$ inherits separability from the quotient $\mathfrak{A}/J$ used in the proof of Theorem 2.1. Hence, a single physically relevant state always leads to a Hilbert space compatible with Axiom I. Theorem 2.2, however, assembles the faithful representation as the direct sum $\mathcal{H} = \bigoplus_{\omega\in\mathcal{F}} \mathcal{H}_\omega$ over an entire separating family $\mathcal{F}$ of states. When inequivalent representations proliferate as in Remark 2.5.3, no countable subfamily of $\mathcal{F}$ typically suffices to separate $\mathfrak{A}$. The resulting direct sum $\mathcal{H}$ is then non separable.

None of this affects the classical construction of Chapter **??** in the same way, since a commutative algebra of continuous functions carries no analogue of an inequivalent representation once its Gelfand spectrum is fixed.

2.6 A Comparison between Classical and Quantum Algebraic Reconstructions

The two classical and quantum constructions share the same starting vocabulary, algebra, states, norm, and representation, yet the order in which these structures arise is different between the two cases. We compare the two constructions, first at the level of the foundational objects and then at the level of the representation theorems that recover a geometric or Hilbert space picture. Particular attention is paid to the bracket. In fact, Chapter **??** had to postulate the Poisson bracket as a structure external to the C^*-algebra, while the present chapter obtained a Lie bracket directly from noncommutativity of the associative product.

Foundational objects

The first difference between the two constructions concerns the order in which the algebraic operations arise. Chapter **??** starts from a fully formed associative commutative product. Differently, the present chapter has to build one from more primitive operations, as the following remark specifies.

Remark 2.6.1 (The associative product, given or constructed). The classical continuous algebra of Definition **??** is an associative commutative algebra of functions under pointwise multiplication. The observable set $\mathcal{O}$ of Definition **??**, in contrast, starts with only rescaling and squaring, and an associative product becomes available only at the end of the whole setting.

The norm, instead, is postulated as part of the function algebra on one side becomes. On the other, it is a consequence of the positivity already required of the states.

Remark 2.6.2 (The norm, given or derived). The supremum norm of Equation (**??**) is inherited directly from the function algebra $C^0(X;\mathbb{C})$ underlying Definition **??**. The norm of Definition **??**, by contrast, is derived from the operational bound on measurement outcomes across all states. Furthermore, the defining property of a C^*-algebra, postulated for the classical case, is a consequence of positivity of the states for the quantum case as shown in Proposition **??**.

Both constructions require their states to separate the algebra, yet the topological and the purely algebraic route to this property follows a completely different path.

Remark 2.6.3 (States and the separating property). Definition **??** characterizes a state, on either side, by positivity and normalization on an already given algebra. In the classical setting, the classical observable algebra $\mathfrak{A}_{sa}$ separates the points of the Gelfand spectrum X as a consequence of Urysohn's lemma, once X is known to be a compact Hausdorff space. In the quantum picture, there is no topological space available yet at the point where separation is needed. Therefore, Definition **??** postulates the separating property as part of the definition of the state space S. It is only afterward, through the family $\mathcal{F}$ of Theorem 2.2, that a faithful representation, and hence an indirect geometric picture, is obtained.

The most concerning difference between the construction is given by the appearance of Lie bracket and commutator. The following proposition isolates the reason why a bracket cannot be recovered from a commutative algebra alone.

Proposition 2.6.4 (The bracket vanishes on an commutative algebra)**.** *Let $\mathfrak{A}$ be an commutative C^*-algebra embedding a Jordan-Banach algebra $\mathcal{O}$ as per Definition* **??**. *Then the bracket $\{A,B\} = \frac{1}{i\hbar}[A,B]$ defined for $A,B \in \mathcal{O}$ vanishes identically.*

Proof. Since $\mathfrak{A}$ is commutative, $[A,B] = AB - BA = 0$ for every $A,B \in \mathfrak{A}$, and in particular for every $A,B \in \mathcal{O} \subset \mathfrak{A}_{sa}$. Hence $\{A,B\} = 0$. □

Remark 2.6.5 (The Poisson bracket, imposed or free). Since $C^0(X;\mathbb{C})$ is commutative, the natural candidate for a bracket built from the commutator is identically zero. Poisson bracket has to be supplied as a new piece of data, namely the smooth dense Poisson subalgebra $\mathfrak{A}^\infty$ of Definition **??**. Hence, the choice of $\mathfrak{A}^\infty$ constitutes the external input of the classical reconstruction, as stated after Definition **??**.

The quantum case follows the opposite path. As soon as $\mathcal{O}$ is realized inside a noncommutative associative algebra through Definition **??**, the commutator $[A,B]$ is generally non-zero. This construction constitutes the way to repair to the innatural choice made in the classical case and to understand that all the structures derived from the Lie algebra follow from noncommutativity.

Representation theorems

Remark 2.3.20 already identifies the classical construction as a special case of the quantum one rather than an analogous procedure. The proposition below specifies this relation.

Proposition 2.6.6 (The classical Gelfand-Naimark theorem as an commutative corollary)**.** *Let $\mathfrak{A}$ be an commutative unital C^*-algebra as per Definition* **??***, and let $\mathcal{F}$ be a separating family of pure states on $\mathfrak{A}$ as per Definition* **??** *and Proposition* **??***. The faithful representation $\pi = \bigoplus_{\omega\in\mathcal{F}} \pi_\omega$ of Theorem 2.2 is unitarily equivalent to the Gelfand isomorphism of Theorem* **??***, under the identification of $\mathcal{F}$ with the Gelfand spectrum $\sigma(\mathfrak{A})$ of Definition* **??**.

For a proof of this statement, see Remark 2.3.20. Consequently $\pi(A)$ coincides with the family $\{\omega(A)\}_{\omega\in\mathcal{F}}$. This relies on results already established, yet its content deserves to be observed independently of the formal statement.

Remark 2.6.7 (Points versus Hilbert spaces). Proposition 2.6.6 shows that the two representation theorems are a single Theorem 2.2, one of them specialized to the commutative case. What changes between the two cases is the meaning attached to a pure state. On one hand, it assumes the meaning of a character as in Definition **??**, evaluating every observable to a real number. On the other hand, it represents a full irreducible representation on a generally infinite-dimensional Hilbert space, per Proposition 2.3.13. Every value $\omega(A)$ that a classical observable takes at a point of X becomes, once commutativity is dropped, an operator $\pi_\omega(A)$ acting on $\mathcal{H}_\omega$, and it is this passage from numbers to operators that the whole of Section 2.3 was devoted to constructing.

The two constructions differ again once the underlying topology enters the argument. Recovering a genuine topological space from the bare algebra is comparatively costly on one side and immediate on the other.

Remark 2.6.8 (Topology, compactness versus completeness). Turning the Gelfand spectrum $\sigma(\mathfrak{A})$ into a compact Hausdorff space, as required by Theorem **??**, needs the weak-$*$ topology of Definition **??** together with the Banach-Alaoglu compactness of Proposition **??**. No analogous step appears in the construction of $\mathcal{H}_\omega$, since the inner product of Equation (2.4) already supplies a metric topology on the quotient $\mathfrak{A}/J$, completed directly into a Hilbert space by the same procedure as Theorem **??**.

Table 2.1 collects every point of comparison drawn in this section into a single overview.

Aspect	**Classical reconstruction, Chapter ??**	**Quantum reconstruction, present chapter**
Associative product	Given from the start, pointwise multiplication on $C^0(X;\mathbb{C})$, Definition **??**	Constructed step by step, from squares to the Jordan product of Definition **??**, to full associativity through the embedding of Definition **??**
Norm	Given, the supremum norm of Equation (**??**)	Derived from the states, $\|A\|^2 = \|A^2\|$ proved in Proposition **??**
Bracket	Imposed externally, smooth dense Poisson subalgebra of Definition **??**, discontinuous in the uniform norm as per Remark **??**	Intrinsic, $\{A,B\} = \frac{1}{i\hbar}[A,B]$ as per Definition **??**, defined on all of $\mathcal{O}$ and proved via Theorem **??**
Separation of states	Via Urysohn's lemma, once X is known to be compact Hausdorff	Postulated directly as part of Definition **??**
Representation theorem	Gelfand-Naimark theorem, Theorem **??**, giving $\mathfrak{A} \cong C_0(X)$	Noncommutative Gelfand-Naimark theorem, Theorem 2.2, via the GNS construction of Theorem 2.1
Topology required	Weak-$*$ topology and Banach-Alaoglu compactness, Definition **??** and Proposition **??**	None beyond the Hilbert space completion of Theorem **??**

Table 2.1: Summary of the comparison between the classical reconstruction of Chapter **??** and the quantum reconstruction of the present chapter.

The comparison collected in Table 2.1 can be summarized in a single statement. Passing from the commutative to the noncommutative case replaces points of a topological space with Hilbert spaces carrying a representation. It also replaces evaluation of a function at a point with the action of an operator on a cyclic vector. Furthermore, it replaces an externally imposed Poisson bracket, confined to a dense subalgebra, with a Lie bracket intrinsic to the whole algebra, as shown in Remark 2.6.5. Every new piece of mathematics required for the noncommutative case, the Jordan product, the GNS construction, and the Lie-Jordan identities of Theorem **??**, exists to compensate for the loss of commutativity and a given geometric space.

2.7 Example II: The Quantum Spin System

This example illustrates the GNS construction of Section 2.3 and the noncommutative Gelfand-Naimark Theorem. The quantum spin is generated by three operators satisfying the $\mathfrak{su}(2)$ commutation relations. In contrast with the classical spin of Section **??** where the same Lie algebra was realised as a Poisson bracket on the sphere. The comparison shows how the same algebraic relations yield a symplectic manifold in the commutative case and a Hilbert space of states in the noncommutative case.

The Pauli Algebra and the GNS Representation

The quantum spin system is modeled by the complex unital C^*-algebra $\mathfrak{A}$, as in Definition **??**, generated by three selfadjoint elements J_1, J_2, J_3 and equipped with the usual C^*-norm and involution. These three generators are required to satisfy the $\mathfrak{su}(2)$ commutation relations

$$[J_i, J_j] = i\,\epsilon_{ijk} J_k, \qquad i, j, k \in \{1, 2, 3\}, \tag{2.43}$$

the same Lie algebra that generated the classical spin system of Section **??** through a Poisson bracket rather than through a commutator. A spin-1/2 system is identified among all the representations of these relations by fixing the value of the Casimir operator $C := J_1^2 + J_2^2 + J_3^2$ to $\frac{3}{4}\mathbb{1}$. Passing to the quotient algebra $\mathfrak{A}_0 := \mathfrak{A}/(C - \frac{3}{4}\mathbb{1})$ then restricts attention to this sector alone, as the analogous restriction did in the classical case of Section **??**. The quotient $\mathfrak{A}_0$ remains a unital C^*-algebra as in Definition **??**. A state ω_0 on $\mathfrak{A}_0$, as in Definition **??**, is fixed by prescribing its values on the identity and on products of at most two generators

$$\omega_0(\mathbb{1}) = 1, \qquad \omega_0(J_i) = 0, \qquad \omega_0(J_i J_j) = \frac{1}{4}\delta_{ij}, \tag{2.44}$$

extended to the whole of $\mathfrak{A}_0$ by linearity. The coefficient $\frac{1}{4}$ appearing in Equation (2.44) is fixed by the Casimir relation itself. In fact, applying ω_0 to the Casimir identity $\sum_i J_i^2 = \frac{3}{4}\mathbb{1}$ and using isotropy yields $3\,\omega_0(J_i^2) = \frac{3}{4}$. It descends that $\omega_0(J_i^2) = \frac{1}{4}$ for every i. The resulting functional is positive and normalized, hence it defines a state on $\mathfrak{A}_0$ by Definition **??**. Because ω_0 treats the three directions J_1, J_2, J_3 symmetrically, it favors none of them, and it is in fact the unique $SO(3)$ invariant state on $\mathfrak{A}_0$. Physically, this symmetry is what identifies ω_0 as the state of complete ignorance about the orientation of the spin.

We carry out this construction explicitly to represent concretly ω_0. It is worth following Remark 2.3.6 in order to keep the space of abstract observables $\mathfrak{A}_0$ on one side and the Hilbert space of state vectors that the construction produces on the other.

1. The first step defines on $\mathfrak{A}_0$ itself the sesquilinear form $(A, B) := \omega_0(A^* B)$, as in Equation (2.4), which is positive semidefinite because ω_0 is a state.

2. The second step isolates the null space of this form, $J := \{A \in \mathfrak{A}_0 \mid \omega_0(B^* A) = 0 \text{ for every } B \in \mathfrak{A}_0\}$, as in Equation (2.5), which coincides, by the Cauchy-Schwarz inequality for $(\cdot, \cdot)$, with the smaller looking set $\{A \in \mathfrak{A}_0 \mid \omega_0(A^* A) = 0\}$ and is a left ideal of $\mathfrak{A}_0$. We identify $\mathfrak{A}_0$ concretely below, showing that this ideal will vanish $J = \{0\}$, so that the algebraic state ω_0 turns out to be faithful.

3. Since $J = \{0\}$, the form $(\cdot, \cdot)$ is already a inner product on the whole of $\mathfrak{A}_0$, and no further quotient is needed before completing it. The *Hilbert space of state vectors* $\mathcal{H}_{\omega_0}$ is obtained by completing $\mathfrak{A}_0$ with respect to the norm induced by this inner product. Note that an element $B \in \mathfrak{A}_0$, which up to this point has only played the role of an abstract observable, is now reinterpreted as a state vector $[B] \in \mathcal{H}_{\omega_0}$. The same mathematical object, an element of $\mathfrak{A}_0$, is thus made to carry two different physical meanings depending on which of the two spaces, $\mathfrak{A}_0$ or $\mathcal{H}_{\omega_0}$, it is regarded as belonging to.

4. The fourth step supplies the *space of physical operators* $\mathcal{B}(\mathcal{H}_{\omega_0})$, the algebra of bounded linear operators on $\mathcal{H}_{\omega_0}$. The representation $\pi_{\omega_0} : \mathfrak{A}_0 \to \mathcal{B}(\mathcal{H}_{\omega_0})$ of Theorem 2.1 is defined by left multiplication. It is through π_{ω_0} that an abstract observable $A \in \mathfrak{A}_0$ is instantiated as the concrete physical operator $\pi_{\omega_0}(A)$. This operator acts on a state vector $[B] \in \mathcal{H}_{\omega_0}$ by

$$\pi_{\omega_0}(A)[B] := [AB], \tag{2.45}$$

as in Equation (2.7). That shows in a single formula the two roles just distinguished, A on the left of the product enters as the observable being represented, while B inside the bracket enters as the state vector it acts on.

5. The fifth and last step identifies the cyclic vector, as per Definition **??**, with $\Psi_{\omega_0} := [\mathbb{1}]$. The state vector corresponding to the identity element of $\mathfrak{A}_0$. This vector closes the construction by reproducing the original algebraic state through the Hilbert space inner product,

$$\left(\Psi_{\omega_0}, \pi_{\omega_0}(A)\Psi_{\omega_0}\right) = \omega_0(A), \qquad \forall A \in \mathfrak{A}_0. \tag{2.46}$$

Hence that the whole of ω_0 is recovered as an expectation value in Ψ_{ω_0}.

The Gelfand-Naimark Theorem and the Operator Algebra

Determining the dimension of $\mathfrak{A}_0$ requires classifying the representations of the $\mathfrak{su}(2)$ Equation (2.43) that are compatible with the fixed value of the Casimir element, $C = \frac{3}{4}\mathbb{1}$. Representation theory of $\mathfrak{su}(2)$ associates the two-dimensional irreducible representation the following

$$\pi_{1/2}(J_i) = \frac{1}{2}\sigma_i. \tag{2.47}$$

In which σ_i denote the Pauli matrices. Fixing the Casimir at $\frac{3}{4}$ forces every $*$-representation of Equation (2.43) to be built out of copies of this single representation alone. This identification of representations translates into an identification of algebras showing that $\mathfrak{A}_0 \cong M_2(\mathbb{C})$. The full argument behind this identification is developed in [5, Section 4.4].

Since the identity together with the three Pauli matrices form a basis of $M_2(\mathbb{C})$, the image of $\pi_{1/2}$ already exhausts the whole of $M_2(\mathbb{C})$. The commutant of $M_2(\mathbb{C})$ inside $\mathcal{B}(\mathbb{C}^2)$ consists only of the scalar multiples of the identity. Schur's Lemma **??** states that a representation is irreducible when its commutant reduces to scalars in this way. Combining the two facts shows that $\pi_{1/2}$ is irreducible. Irreducibility of $\pi_{1/2}$ has an immediate consequence for the states it defines. By Proposition 2.3.13, every vector state

$$\omega_\psi(A) = \left(\psi, \pi_{1/2}(A)\psi\right), \qquad \forall\, \psi \in \mathbb{C}^2, \tag{2.48}$$

is pure, and $\pi_{1/2}$ is, up to unitary equivalence, the unique irreducible representation of $\mathfrak{A}_0$.

Having established an irreducible representation of $\mathfrak{su}(2)$, we continue by write the state ω_0 analytically. The basis $\{\mathbb{1}, \sigma_1, \sigma_2, \sigma_3\}$ of $M_2(\mathbb{C})$ satisfies $\mathrm{Tr}(\mathbb{1}) = 2$, $\mathrm{Tr}(\sigma_i) = 0$, and $\mathrm{Tr}\left(\sigma_i\sigma_j\right) = 2\delta_{ij}$. These are the values of Equation (2.44), up to a factor $\frac{1}{2}$, used to define ω_0, and by linearity the two functionals coincide on the whole of $\mathfrak{A}_0$,

$$\omega_0(A) = \frac{1}{2}\,\mathrm{Tr}(A), \qquad \forall A \in \mathfrak{A}_0. \tag{2.49}$$

The trace of a strictly positive operator on $\mathbb{C}^2$ is itself strictly positive, that is $\omega_0(A^*A) = \frac{1}{2}\,\mathrm{Tr}(A^*A) > 0$ for every non-zero $A \in \mathfrak{A}_0$. The state ω_0 is therefore a faithful state by Definition 2.3.17. This confirms the vanishing of the left kernel anticipated in the second step above, $J = \{0\}$. The construction above then yields

$$\mathcal{H}_{\omega_0} \cong \mathfrak{A}_0 \cong \mathbb{C}^4. \tag{2.50}$$

This identification of $\mathcal{H}_{\omega_0}$ with $\mathbb{C}^4$ has to be read with care, since the four-dimensional space $\mathbb{C}^4$ appearing here does not describe a new physical system with four levels. It is instead $M_2(\mathbb{C})$ of all linear operators on $\mathbb{C}^2$, now regarded as a Hilbert space of state vectors rather than as an algebra of observables because it includes also density matrices, as in Definition 2.3.15.

Faithfulness of ω_0 has another consequence, it makes the representation π_{ω_0} itself injective. Consequently, Proposition 2.3.4 shows that an injective representation is isometric. That is the property that the Gelfand-Naimark Theorem of Section 2.3 guarantees in general. Making π_{ω_0} explicit is the remaining task of this subsection.

To achieve that, we first recall the canonical vector space isomorphism $M_2(\mathbb{C}) \cong \mathbb{C}^2 \otimes \mathbb{C}^2$. Let $\{e_1, e_2\} \in \mathbb{C}^2$ be the standard orthonormal basis of $\mathbb{C}^2$, and let $E_{ij} \in M_2(\mathbb{C})$ denote the matrix units, defined by their action $E_{ij}e_k = \delta_{jk}e_i$. The isomorphism $M_2(\mathbb{C}) \cong \mathbb{C}^2 \otimes \mathbb{C}^2$ is fixed by the linear extension of the map $E_{ij} \mapsto e_i \otimes e_j$. Then, every abstract observable of $\mathfrak{A}_0$ acquires, once identified with its matrix representation, a corresponding vector in $\mathbb{C}^2 \otimes \mathbb{C}^2$.

Under the isomorphism just fixed, left multiplication acts only on the first tensor factor. For any observable $A \in \mathfrak{A}_0$, with matrix representation $\pi_{1/2}(A) \in M_2(\mathbb{C})$,

$$\pi_{1/2}(A)\,E_{ij} \longleftrightarrow \big(\pi_{1/2}(A) \otimes \mathbb{1}_{\mathbb{C}^2}\big)(e_i \otimes e_j). \tag{2.51}$$

By contrast, right multiplication by an arbitrary matrix $B \in M_2(\mathbb{C})$ acts only on the second tensor factor,

$$E_{ij}\,B \longleftrightarrow \big(\mathbb{1}_{\mathbb{C}^2} \otimes B^T\big)(e_i \otimes e_j), \tag{2.52}$$

where B^T denotes the transpose of B with respect to the basis $\{e_1, e_2\}$. We observe that the first factor carries the physical degrees of freedom of the spin. In fact, it is on this factor alone that the observable algebra acts through π_{ω_0}. Given that, for every $B \in M_2(\mathbb{C})$, the operators $\mathbb{1}_{\mathbb{C}^2} \otimes B^T$ commute with every operator $\pi_{1/2}(A) \otimes \mathbb{1}_{\mathbb{C}^2}$ with $A \in \mathfrak{A}_0$, the second factor is what generates the commutant of the observable algebra inside $\mathcal{B}(\mathcal{H}_{\omega_0})$. Therefore, we can write the representation π_{ω_0} in the following way

$$\pi_{\omega_0}(A) = \pi_{1/2}(A) \otimes \mathbb{1}_{\mathbb{C}^2}, \qquad A \in \mathfrak{A}_0. \tag{2.53}$$

The cyclic vector, as in Definition **??**, is identified above as $\Psi_{\omega_0} = [\mathbb{1}]$. The identity matrix decomposes as $\mathbb{1} = E_{11} + E_{22}$. Under the map shown earlier, we can express the cyclic vector as

$$\Psi_{\omega_0} \longleftrightarrow \frac{1}{\sqrt{2}}\big(e_1 \otimes e_1 + e_2 \otimes e_2\big). \tag{2.54}$$

This vector embodies the mechanism of purification. The state ω_0, which on $\mathbb{C}^2$ alone requires the density matrix $\rho = \frac{1}{2}\mathbb{1}$ as in Equation (2.20), is here represented instead as a vector state in the enlarged space $\mathbb{C}^2 \otimes \mathbb{C}^2$. That is engineered so that its expectation value on every physical observable $\pi_{1/2}(A) \otimes \mathbb{1}_{\mathbb{C}^2}$ reproduces the evaluation already dictated by the original density matrix $\rho = \frac{1}{2}\mathbb{1}$.

The Bloch Sphere and Mixed States

The pure states of the system are the rays of $\mathbb{C}^2$. That is the equivalence classes of unit vectors modulo multiplication by a phase. The space of such rays is the complex projective line $\mathbb{PC}^1$, subject of work in the next Chapter. Each vector $\psi \in \mathbb{C}^2$ identifies a ray $[\psi] \in \mathbb{PC}^1$. That is parametrised by the expectation values of the Pauli matrices given by $\mathbf{r} = (\psi, \boldsymbol{\sigma}\psi) \in \mathbb{C}^3$, which satisfy $|\mathbf{r}| = 1$ to determine the ray uniquely. Therefore $\mathbb{PC}^1 \cong \mathbb{S}^2$, and this identification is called the Bloch sphere. States on this projective space are defined via

$$\omega_\psi(A) = \big(\psi, \pi_{1/2}(A)\psi\big), \qquad \forall\, \psi \in \mathbb{C}^2,\ \|\psi\| = 1. \tag{2.55}$$

More generally, every state of $\mathfrak{A}_0$ is

$$\omega_\rho(A) = \operatorname{Tr}\big(\rho\, \pi_{1/2}(A)\big), \qquad \rho = \tfrac{1}{2}(\mathbb{1} + \mathbf{r}\cdot\boldsymbol{\sigma}),\ \ |\mathbf{r}| \le 1, \tag{2.56}$$

and its space is called the Bloch ball. Pure states are the boundary $|\mathbf{r}| = 1$, where ω_0 is the centre $\mathbf{r} = 0$ representing the maximally mixed state.

By Theorem 2.1, every state, pure or mixed, has a *cyclic* GNS vector. What changes with $\mathbf{r}$ is the irreducibility of the representation for Proposition 2.3.13. For $|\mathbf{r}| < 1$, ρ is invertible and ω_ρ is faithful, so the argument used above for ω_0 applies. Only at $|\mathbf{r}| = 1$ does it collapse to the irreducible $\pi_{1/2}$ on $\mathbb{C}^2$.

Comparison with the Classical Spin System

The classical spin of Section **??**, generated by x_1, x_2, x_3 with $\{x_i, x_j\} = \epsilon_{ijk}x_k$ yields the commutative algebra $C(\mathbb{S}^2)$. Its Gelfand spectrum, which was the sphere itself, is a continuum of pure states. The quantum spin uses the same Lie algebra noncommutatively admitting a *unique* irreducible representation $\pi_{1/2}$ up to unitary equivalence. Its pure states reproduce the same sphere $\mathbb{S}^2$. But the full quantum state space, the Bloch ball, is strictly richer. In fact, it contains mixed states with no counterpart in $C(\mathbb{S}^2)$. The GNS construction and the Gelfand-Naimark Theorem are what let the same $\mathfrak{su}(2)$ algebra generate a symplectic manifold commutatively and a Hilbert space noncommutatively.

These extra features arise from the difference in the *geometry* of the two state spaces. The classical state space, representing probability measures on $\mathbb{S}^2$, as show in Section **??**, is a generally simplex. Moreover, every classical mixed state decomposes uniquely into pure ones. However, the Bloch ball is strictly convex, and this uniqueness fails completely. This makes ω_0 at its centre an equal mixture of ω_ψ and $\omega_{\psi^\perp}$ for *every* unit vector $\psi \in \mathbb{C}^2$. A classical mixed state can always be read as an ignorance about which pure state actually obtains. Instead, a generic quantum mixed state cannot be given such a reading at all. It is this loss of a unique "ignorance interpretation" that foundational frameworks take as the operational signature of nonclassicality.

The next Example **??** will continue this formulation leading the discussion to time evolution, symmetries and Quantum Noether's Theorem **??** that will be presented in the next chapter.

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